Real-Time SAR Ship Detection: A Speckle-Model-Based DSP Pipeline

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Outline

- Introduction and Motivation
- Evaluation Framework
- SAR Echo and Imaging Model
- Sparse Image Refinement
- Experimental Results
- Conclusion
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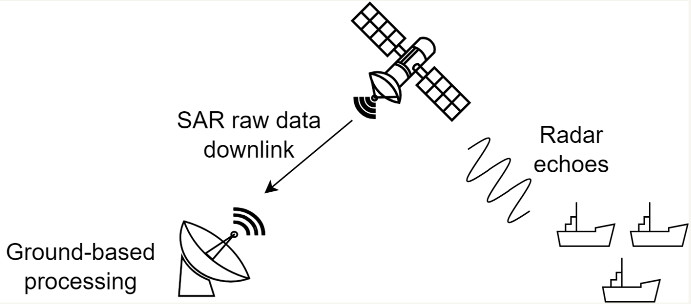
Introduction

- SAR raw data are downlinked for ground-based processing, including SAR focusing, image enhancement, and ship detection.
- Detected ships provide information for maritime surveillance.



Problem and Motivation

- Ground-Based Processing
  - Large raw-data volume
  - Limited downlink bandwidth
  - Delayed target information
- Onboard Processing
  - Reduced data transmission and detection latency
  - Challenge: Limited computation, memory, and power
- Research Goal
  - Lightweight SAR processing for low-latency ship detection

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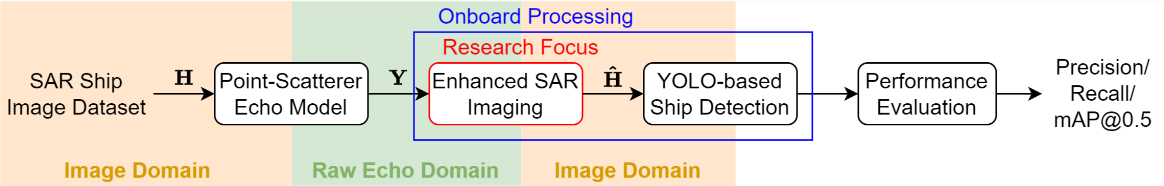
Evaluation Framework

- Research Focus: Enhanced SAR imaging
- Evaluation Task: YOLO-based ship detection

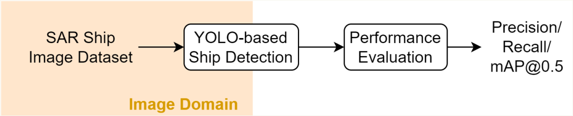


- Evaluation Metrics [1]
  - Precision: $\frac{TP}{TP+FP}$
  - Recall: $\frac{TP}{TP+FN}$
  - mAP@0.5: Mean AP across classes at IoU threshold = 0.5

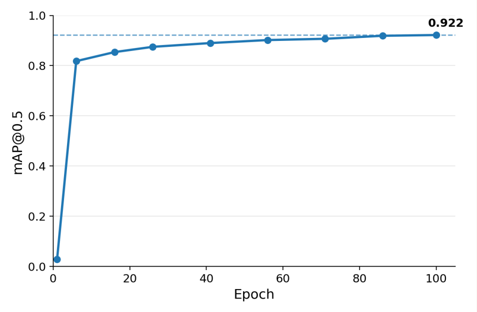
TP: True Positive, FP: False Positive, FN: False Negative

Dataset and Detection Model Setup

- HRSID [2]
  - High-Resolution SAR Images Dataset
  - Training: 3,642 images
  - Validation: 1,962 images
  - Image size: 800 $\times$ 800 pixels
- YOLO Detection Model
  - YOLO11m fine-tuned on HRSID (100 epochs)



- Validation Performance
  - Precision: 0.910
  - Recall: 0.840
  - mAP@0.5: 0.922

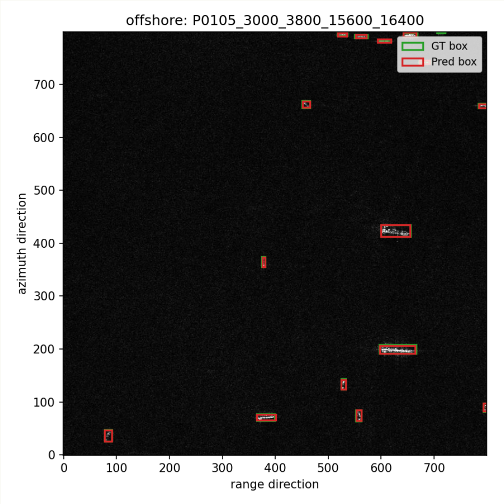


YOLO Ship Detection Results (1)

	Overall	Inshore	Offshore
Images	1,962	369	1,593
Precision	0.913	0.803	0.982
Recall	0.846	0.741	0.965
mAP@0.5	0.918	0.817	0.984

- Offshore scenes achieve better detection performance than inshore scenes [3].

YOLO Ship Detection Results (2)



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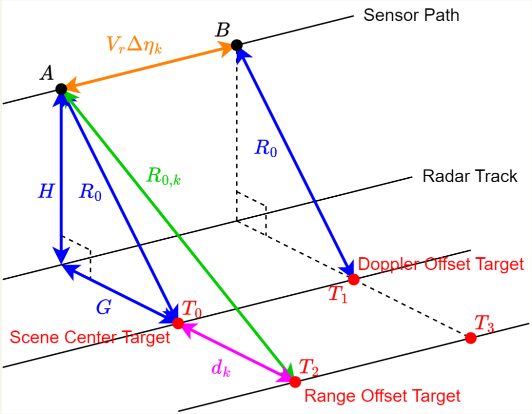
Point-Scatterer Echo Model: Geometry



- Target position determines the time-varying slant range $R_k(\eta_m)$.
- $R_k(\eta) = \sqrt{R_{0,k}^2 + (\eta_m + \Delta\eta_k)^2 V_r^2}$
  - $R_{0,k} \approx R_0 + d_k$
  - R_0: slant range to scene center
  - d_k: ground-range offset of scatterer k, approximated as slant-range offset
  - $\eta_m = m\Delta\eta$: slow time
  - $\Delta\eta_k$: azimuth-time offset of scatterer k

Point-Scatterer Echo Model: Signal Synthesis

- Geometry-Dependent Delay and Phase
  - $\tau_k(\eta_m) = \frac{2R_k(\eta_m)}{c}$, $\phi_k(\eta_m) = -4\pi\frac{R_k(\eta_m)}{\lambda}$
- Echo Synthesis: $\mathbf{S} = [s_{m,n}]_{M\times N}$

$$s_{m,n} = \sum_k A_k \,\mathrm{rect}\left(\frac{\tau_n - \tau_k(\eta_m)}{T_r}\right) \cdot e^{j\pi K_r(\tau_n - \tau_k(\eta_m))^2} \cdot e^{j\phi_k(\eta_m)}$$

  - $\eta_m = m\Delta\eta$: slow time, $\tau_n = n\Delta\tau$: fast time
  - A_k: scattering coefficient of scatterer k, $K_r = \frac{B_r}{T_r}$

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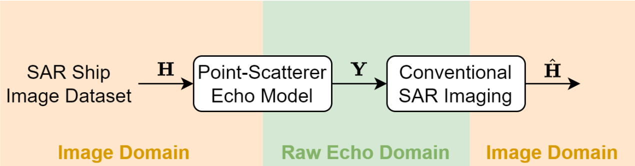
Conventional SAR Imaging



- Imaging goal: Recover **H** from **Y**.
- Conventional methods: RDA, CSA, and BP

One-Step Sparse Image Refinement

- Image-domain regularization [4]

$$\hat{\mathbf{H}} = \arg\min_{\mathbf{H}} \underbrace{\frac{1}{2}\|\mathbf{H} - \mathbf{H}_{\text{CSA}}\|_F^2}_{\text{Preserve CSA image}} + \lambda \underbrace{\|\mathbf{H}\|_1}_{\text{Promote sparsity}}$$

  - Strong scatterers $\rightarrow$ retained; weak components $\rightarrow$ suppressed
  - $\hat{\mathbf{H}} = \mathcal{S}_\lambda(\mathbf{H}_{CSA})$, $\mathcal{S}_\lambda(x) = \frac{x}{|x|}\max(|x| - \lambda, 0), x \neq 0$
- One-Step Refinement
  - $\mathbf{Y} \xrightarrow{\text{CSA}} \mathbf{H}_{\text{CSA}} \xrightarrow{\mathcal{S}_\lambda(.)} \hat{\mathbf{H}}$

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Simulation Setup

- Dataset: 100 offshore SAR images sampled from 1,593 images
- Evaluation: Ship detection on reconstructed SAR images
- SAR Simulation Parameters

Parameter	Value	Parameter	Value
Pulse Width	12.5 μs	Wavelength	0.1152 m
Pulse Repetition Frequency	1 kHz	Closest Slant Range	4000 m
Bandwidth	50 MHz	Platform Velocity	120 m/s
Sampling Frequency	64 MHz	Antenna Length	1.2 m

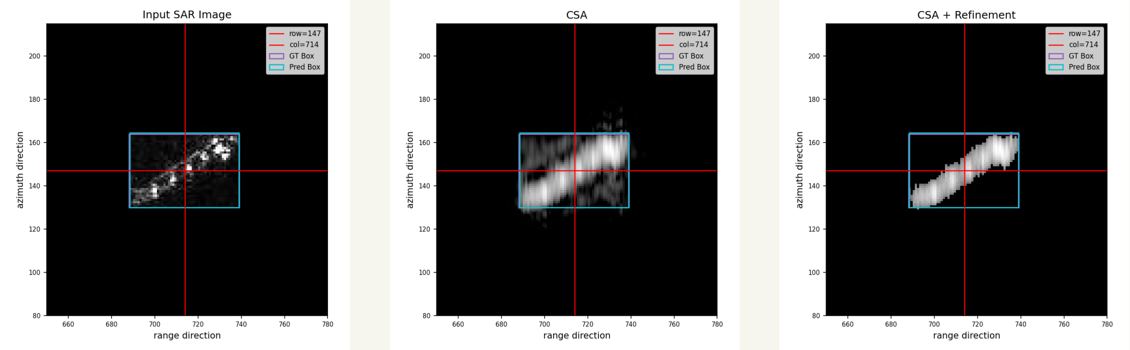
Ship Detection Performance

Image	Precision	Recall	mAP@0.5
Input SAR Image	1.000	0.978	0.975
CSA	0.977	0.940	0.954
CSA + Refinement	0.993	0.986	0.985

- Thresholding suppresses CSA-induced background artifacts and improves ship detection performance.

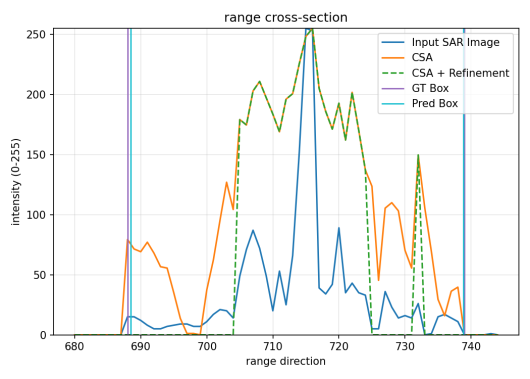
Representative Example

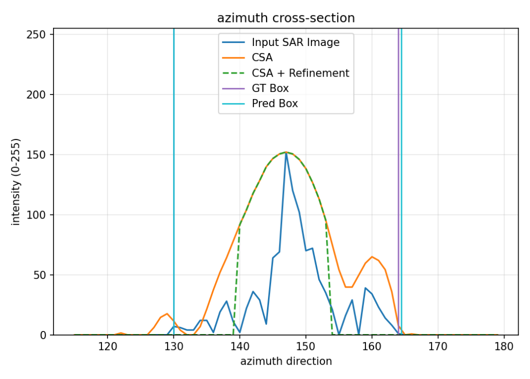
- Observation: Thresholding suppresses background artifacts while preserving the dominant ship response.



Cross-Section Analysis

- Thresholding removes low-amplitude sidelobe/background responses while retaining the dominant target structure.





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Conclusion

- Established an end-to-end framework for evaluating SAR image enhancement through ship detection.
- One-step sparse refinement suppresses background artifacts while preserving dominant ship responses.
- CSA + refinement improves detection performance from 0.954 to 0.985 mAP@0.5.

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[1] F. Song, S. Yang and X. Zhou, “SARES-DEIM: Sparse Mixture-of-Experts Meets DETR for Robust SAR Ship Detection,” in *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing (JSTARS)*, 2026.

[2] S. Wei, X. Zeng, Q. Qu, M. Wang, H. Su and J. Shi, “HRSID: A High-Resolution SAR Images Dataset for Ship Detection and Instance Segmentation,” in *IEEE Access*, vol. 8, pp. 120234–120254, 2020.

[3] T. Zhang *et al.*, “Balance Scene Learning Mechanism for Offshore and Inshore Ship Detection in SAR Images,” in *IEEE Geoscience and Remote Sensing Letters*, vol. 19, pp. 1–5, 2022.

[4] H. Bi, G. Bi, B. Zhang, W. Hong and Y. Wu, “From Theory to Application: Real-Time Sparse SAR Imaging,” in *IEEE Transactions on Geoscience and Remote Sensing*, vol. 58, no. 4, pp. 2928–2936, April 2020.